Consensus-Supervised U-Nets with PQ-Optimal Instance Selection and Refinement for Solar Filament Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Ten U-Nets (EfficientNetV2-S and EfficientNet-B3 encoders, five date-grouped folds each) are trained at native resolution with *soft consensus* targets, the fraction of annotators who marked each pixel. Their averaged probability map is converted into instances in two learned stages. A decision layer, adapted from a public competition notebook, thresholds the map at 16 levels, scores every nested connected component with a gradient-boosted model of its expected matched IoU against a random annotator, and selects the pixel-disjoint subset that maximises the competition's pooled Panoptic Quality (PQ) with an exact dynamic programme. A second U-Net then re-segments every selected instance from a local window. On all 707 training images the out-of-fold pooled PQ rises from 0.405 (threshold and minimum area) to 0.447, with a gain on every fold, and the public leaderboard score from 0.35 to 0.39; two annotators scored against each other reach 0.34. Code, trained weights and a notebook reproducing the submission are public.[1]

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[2]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

The task is *instance* segmentation evaluated with Panoptic Quality [9]: every predicted filament counts either as a true positive (IoU $>$ 0.5 with a ground-truth filament) or as a false positive, and each pixel may belong to at most one prediction. Three properties of the data shaped our design. First, the 707 training images carry 1154 annotation sets, because 296 images were annotated independently by two or three annotators, and the annotators disagree strongly: scoring one annotator's outlines against another's on these images gives a pooled PQ of only 0.340 (mean matched IoU 0.63, recognition quality 0.54). Second, filaments are small and thin (median area 1228 px in a 2048×2048 frame), so down-sampling destroys the barbs the organizers care about. Third, only 0.9% of the 8199 annotated filaments touch another filament within 3 px, so the problem is almost entirely semantic segmentation followed by a decision about *which* connected regions to submit and *how far* each extends. We therefore pair semantic U-Nets [10] with two learned, metric-driven stages, and measure every design choice on out-of-fold data.

[1] https://github.com/cloudforge-oy/solar-filament-2026
[2] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

2.1 Data preparation and targets

Images are read as single-channel 8-bit grayscale; no limb-darkening correction or disk masking is applied. For every annotation set the polygons are rasterised with `pycocotools`, which reproduces the `area` field of the annotation file exactly. The k annotation sets of an image are reduced to three *consensus* targets: (i) the soft filament mask $y_{\text{seg}} \in [0, 1]$, the fraction of annotators who marked the pixel, so that the network predicts the *expected* annotation; (ii) a normalised distance-to-boundary map, averaged over annotators; (iii) a contact-line map of pixels within 2 px of two instances of one annotation set. The two auxiliary targets were meant to separate touching filaments; neither a distance-map watershed [4] nor a contact-line cut ever improved validation PQ, so they are kept as auxiliary tasks whose predicted maps serve as candidate descriptors (Section 2.3).

2.2 Dense models and training

We use the U-Net of `segmentation_models_pytorch` [8] with ImageNet-pretrained encoders, one input channel and three output channels, trained with $\mathcal{L}$ = BCE + Dice on the soft mask, 5 MSE on the distance map and 0.5 $\text{BCE}_{w=20}$ on the contact map. Two encoder families are averaged: EfficientNet-B3 [11] (640-px crops, batch 6, 50 epochs, inputs standardised with dataset constants) and EfficientNetV2-S [12] (512-px crops, batch 16, 75 epochs, inputs standardised with each image's own mean and deviation inside the solar disk). Crops are taken at native resolution, four per image per epoch, centred near a filament pixel with probability 0.7; augmentations are flips, 90° rotations, gamma, contrast, brightness and noise. Optimisation uses AdamW (weight decay 10^{-4}), a one-cycle schedule peaking at $3 \cdot 10^{-4}$ and mixed precision; one fold takes 45–50 minutes on an RTX 4080.

Validation split. The 707 images are grouped by observation date and the dates are assigned to five folds, so that the same day observed by two GONG stations, or the same image annotated by three annotators, never straddles the training/validation boundary. Every model is trained once per held-out fold, so each of the 1154 annotation sets receives an out-of-fold prediction; the two later

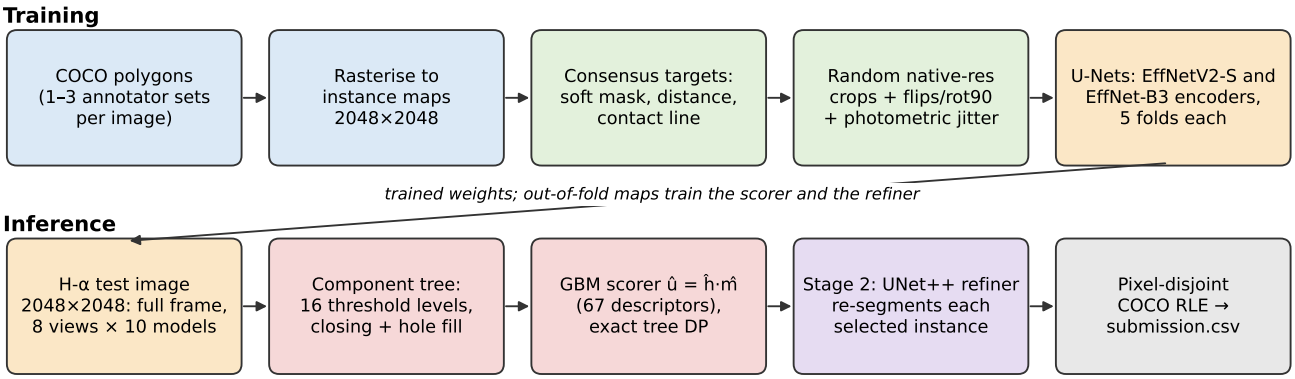


Figure 1: End-to-end pipeline. Training (top): COCO polygons of all annotator sets are rasterised into instance maps and reduced to three consensus targets per image; U-Nets are trained on random native-resolution crops. Inference (bottom): full-frame prediction averaged over eight dihedral views and ten models, a 16-level component tree, a learned score per candidate with an exact selection that maximises pooled PQ, instance-wise refinement, COCO RLE encoding.

stages are fitted on these out-of-fold maps with the same folds (*cross-fitting*), and all numbers in Section 3 are out-of-fold end to end.

2.3 PQ-optimal instance selection

Each image is predicted as a whole 2048×2048 frame. The prediction is averaged over eight views (flips and 90° rotations) and over all models, which gives one probability map p per image. Turning p into filaments is a decision problem: which regions do we submit, and where do we cut each one? We follow the idea of a public competition notebook [13] and re-implemented it on our own maps (`src/decision.py`).

What the metric rewards. The host sums true positives, false positives, false negatives and matched IoUs over all annotator records. For a candidate region c in an image with R records, let

$$u(c) = \frac{1}{R} \sum_{r} \mathrm{IoU}(c, g_r)\, \mathbf{1}\{\mathrm{IoU}(c, g_r) > 0.5\}$$

be its matched IoU averaged over the annotators, where g_r is the best-matching filament of record r. Every submitted region adds $u(c)$ to the numerator of PQ and one half to the denominator, whether it matches or not. A region is therefore worth submitting only if $u(c)$ exceeds a fixed threshold τ, and for the best possible submission τ equals half of its PQ [5]. In short: submit the pixel-disjoint set of regions with the largest $\sum_c (u(c) - \tau)$.

Candidates. We threshold p at 16 levels from 0.05 to 0.80, close small gaps and fill holes. The connected regions of all levels (at least 100 px) are the candidates, about 176 per image. A region at a higher level always lies inside one region of the level below, so the candidates form a tree, and any choice of regions that are not nested in each other is a valid submission without overlaps.

Scorer. The true u is unknown at test time, so we predict it. For every candidate we compute 67 descriptors from the image and the predicted maps only: threshold level, size and shape, statistics of p and of the auxiliary maps, darkness against the surroundings, position on the disk, the mean probability of each encoder family, and where the region sits in its tree (how many levels it survives, how it splits, its parent and siblings, its rank in the image). No information from the file name is used. Gradient-boosted trees [6] predict two things: the probability that an annotator record matches the region, and the IoU if it does. Their product is the predicted $\hat{u}$. We average four such scorers trained with different random seeds.

Selection. Given $\hat{u}$, the best set of non-nested regions is found exactly by one pass over the tree from the leaves up: for each region we keep the better of taking the region itself or taking the best choice among its children. τ is set to the middle of the flat top of the validation curve (Figure 2). One correction is needed at test time. The scorer is trained on maps from a single model per encoder family, while test maps average five models and are smoother, so the same τ keeps fewer regions. We therefore choose τ on the test maps, without labels, so that the number of selected regions per image equals the validated value (6.7).

2.4 Instance refinement

p estimates the probability that a *random* annotator marks a pixel. For a filament that only some annotators draw, p is low everywhere inside it and no level set gives the outline an annotator would draw. The second stage (`src/refine.py`) is a UNet++ [14] with a ResNet34 encoder [7] that sees a native-resolution window around *one* selected instance (four channels: image, p, the instance, the other selected instances) and predicts $q(x)$, the probability that x lies inside the annotator's outline given that the annotator drew this filament. Its target is the mean of the ground-truth outlines matched to the seed (IoU > 0.5). Training seeds are the out-of-fold selections and near-selections of the decision layer (≈86 per image, drawn 12 000 per epoch for ten epochs on 256-px windows); the refiner of fold k never sees fold-k images. At inference the window is the instance's bounding box plus 48 px, q is averaged over four

flips and the five fold models, and the refined mask is the part of $\{q > 0.45\}$ connected to the seed. It may not enter another selected instance, instances take pixels in descending score order, and a refined mask whose area leaves $[0.4, 3]$ times the seed area falls back to the seed, so the submission stays pixel-disjoint.

3 EVALUATION

3.1 Strategy

We re-implemented PQ as defined by the organizers: predicted and ground-truth instances are matched by IoU > 0.5 against every annotation set of an image, and TP, FP, FN and matched IoUs are *pooled* over all 1154 (image, record) pairs as the host does. Every number below is out-of-fold for the dense models, the scorer and the refiner at once.

3.2 Results

Table 1 lists the steps in the order they were added, and Table 2 the final pipeline per fold. The plain rule (components of $p > 0.6$ with area ≥ 400) and the best single tree level score the same (0.405 and 0.407), so the gain of the decision layer does not come from the closing or the extra levels: it keeps fewer, better candidates (false positives fall from 4153 to 2558 for 108 fewer true positives on the final maps). Tree-context descriptors are the most useful addition to the scorer (+0.006); geometry, neighbourhood and scorer hyperparameters each change PQ by less than 0.002. A second encoder family raises the ceiling of the candidate trees (selection with the true u: 0.559 to 0.568) but the realised score by only 0.001. Two descriptors read from the file name (observatory and year) were removed to comply with the hosts' rule that inference may not use observation metadata; this costs 0.002. The refiner adds +0.007 and improves every fold. Trained on an earlier selection, UNet++ scored 0.449, a plain U-Net 0.446 and their average 0.448, so refiner variants differ by no more than run-to-run variation.

Table 1: Pooled out-of-fold PQ (707 images) and public leaderboard score as components were added. Rows without a leaderboard entry were not submitted on their own.

Pipeline	OOF PQ	public LB
EfficientNet-B3, flip TTA, plain rule	0.405	0.35
+ decision layer (35 descriptors)	0.431	0.37
+ 8-view TTA, tree-context descriptors	0.439	0.38
+ averaged scorers	0.441	–
+ EfficientNetV2-S family	0.442	–
– file-name descriptors (site, year)	0.440	–
+ instance refiner (final)	**0.447**	**0.39**

What did not help. (i) Averaging more dense models beyond the two families. (ii) Letting a candidate group fragments at a low threshold while keeping the tighter shape of a high one: the oracle ceiling does not move (0.558 on fold 0). (iii) A CNN that scores each candidate from its image window: with early stopping it adds $+0.0004 \pm 0.0015$; over-trained it adds +0.003, but only on images within three days of a training image, i.e. by memorising neighbouring-day labels, so we left it out.

Table 2: Final pipeline per fold, pooled out-of-fold PQ: plain rule, decision layer, decision layer with refiner, and the selection with the true u (ceiling of the candidate trees without refinement).

Fold	plain rule	decision layer	+ refiner	Δ	oracle u
0	0.415	0.441	**0.450**	+0.035	0.569
1	0.390	0.436	**0.444**	+0.054	0.563
2	0.422	0.451	**0.459**	+0.037	0.575
3	0.400	0.433	**0.442**	+0.042	0.566
4	0.401	0.438	**0.442**	+0.042	0.571
all	0.405	0.440	**0.447**	+0.042	0.568

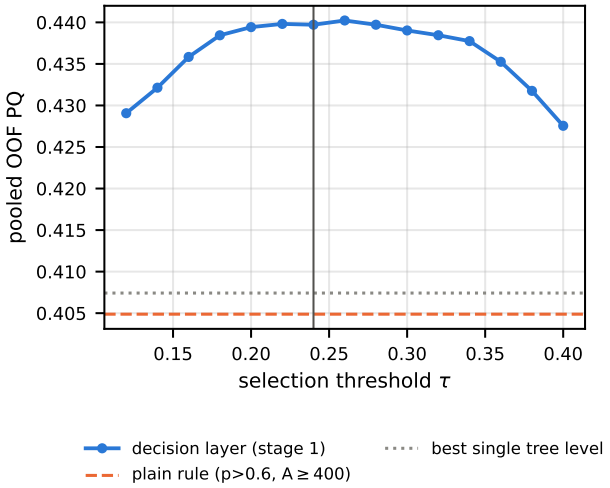


Figure 2: Pooled out-of-fold PQ of the decision layer against the selection threshold τ (cross-fitted scorer), with the plain rule and the best single tree level. The vertical line marks the centre of the plateau.

3.3 Discussion

Strengths. Every stage optimises the competition metric directly and is validated without leakage across folds; the submission is pixel-disjoint by construction; inference takes about 17 s per image on a laptop GPU.

Limits. The score is bounded by annotator disagreement rather than by image evidence: our out-of-fold PQ (0.447) is already well above the agreement between two annotators (0.340), and with the true u the candidate trees would reach 0.57, so most of the remaining loss is in guessing which faint filaments a particular annotator draws and how they group fragments. Every honest pipeline we could inspect, including ours, loses $\approx$0.06 between out-of-fold and public leaderboard PQ, for models of very different strength, which points to the labels of the public test images rather than to overfitting. Public leaderboard scores above 0.5 should be read with care: the published scorer counts every prediction that overlaps a ground-truth filament by IoU > 0.5, so repeating a mask adds true

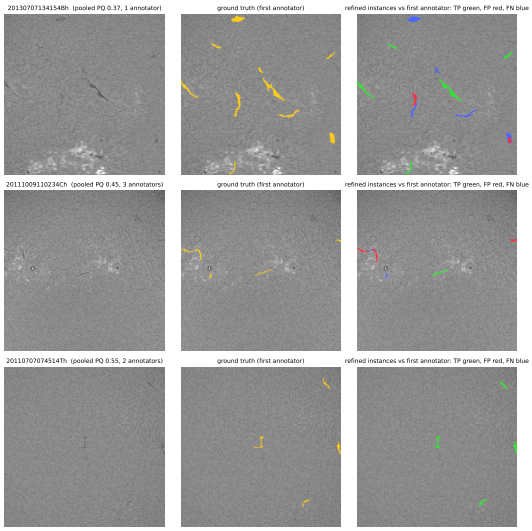


Figure 3: Fold-0 validation images at the 25th, 50th and 75th percentile of per-image pooled PQ (out-of-fold maps, scorer and refiner). Left: H-α crop; middle: the first annotator's ground truth; right: refined instances coloured against that annotator (true positive green, false positive red, missed filament blue).

positives [3]; applied to our own out-of-fold predictions, duplication alone would move 0.44 to 0.52–0.66. All our submissions are pixel-disjoint, as the rules require.

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